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GATE ECE · SIGNALS AND SYSTEMS · CHAPTER-WISE PYQS

Sampling

GATE ECE Signals and Systems · Previous Year Questions with Answer Key

31	27	1988 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

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Marks 1	13 questions
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Answer key	at the end

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Marks 1 13 questions

Q1. GATE ECE 2026

Two analog signals $x_1(t)$ and $x_2(t)$ (t in second), are sampled at a rate $F_s = 40$ Hz, where $x_1(t) = \cos(20\pi t), t \geq 0$, and $x_2(t) = \cos(100\pi t), t \geq 0$.

The first ten samples (starting from $t = 0$) are considered for the analysis.

Which of the following statements is TRUE?

(A) All of the first three samples of $x_1(t)$ are greater than the corresponding samples of $x_2(t)$.

(B) All of the last three samples of $x_1(t)$ are greater than the corresponding samples of $x_2(t)$.

(C) All of the samples of $x_2(t)$ are greater than the corresponding samples of $x_1(t)$.

(D) All of the fourth to seventh samples of $x_1(t)$ are equal to the corresponding samples of $x_2(t)$.

Open solution

Q2. GATE ECE 2016 Set 1

A continuous-time function $x(t)$ is periodic with period T. The function is sampled uniformly with a sampling period T_s. In which one of the following cases is the sampled signal periodic?

(A) $T = \sqrt{2}\, T_s$ (B) $T = 1.2T_s$ (C) Always (D) Never

Open solution

Q3. GATE ECE 2016 Set 3

Consider the signal $x(t) = \cos(6\pi t) + \sin(8\pi t)$, where t is in seconds. The Nyquist sampling rate (in samples/second) for the signal $y(t) = x(2t + 5)$ is

(A) 8 (B) 12 (C) 16 (D) 32

Open solution

Q4. GATE ECE 2015 Set 2

The signal $\cos\left(10\pi t + \frac{\pi}{4}\right)$ is ideally sampled at a sampling frequency of 15 Hz. The sampled signal is passed through a filter with impulse response $\left(\frac{\sin(\pi t)}{\pi t}\right)\cos\left(40\pi t - \frac{\pi}{2}\right)$. The filter output is

(A) $\frac{15}{2}\cos\left(40\pi t - \frac{\pi}{4}\right)$

(B) $\frac{15}{2}\left(\frac{\sin(\pi t)}{\pi t}\right)\cos\left(10\pi t + \frac{\pi}{4}\right)$

(C) $\frac{15}{2}\cos\left(10\pi t - \frac{\pi}{4}\right)$

(D) $\frac{15}{2}\left(\frac{\sin(\pi t)}{\pi t}\right)\cos\left(40\pi t - \frac{\pi}{2}\right)$

Open solution

Q5. GATE ECE 2014 Set 1

Consider two real valued signals, $x(t)$ band - limited to $[-500Hz,\ 500Hz]$ and $y(t)$ band - limited to $[-1\ kHz,\ 1kHz]$. For $z(t) = x(t).y(t)$, the Nyquist sampling frequency $(in\ kHz)$ is_____.

Open solution

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Q6. GATE ECE 2014 Set 3

Let $x\ (t)\ =\ \cos\ (10\pi t)\ +\ \cos\ (30\pi t)$ be sampled at $20\ Hz$ and reconstructed using an ideal low-pass filter with cut-off frequency of $20\ Hz$. The frequency/frequencies present in the reconstructed signal is/are.

Open solution

(A) 5 Hz and 15 Hz only
(B) 10 Hz and 15 Hz only
(C) 5 Hz, 10 Hz and 15 Hz only
(D) 5 Hz only

Q7. GATE ECE 2014 Set 3

A modulated signal is $y\,(t)\ =\ \ \ m\,(t)\,\cos\,(40000\pi t)$, where the baseband signal $m\,(t)$ has frequency components less than 5 kHz only. The minimum required rate (in kHz) at which $y\ (t)$ should be sampled to recover $m\ (t)$ is _____.

Open solution

Q8. GATE ECE 2013

A band-limited signal with a maximum frequency of 5 kHz is to be sampled. According to the sampling theorem, the sampling frequency which is not valid is

(A) 5 kHz
(B) 12 kHz
(C) 15 kHz
(D) 20 kHz

Open solution

Q9. GATE ECE 2002

Consider a sampled signal $y\,(t) = 5 \times 10^{-6}\,x\,(t)\ \sum_{n=-\infty}^{+\infty}\delta\,(t - nT_s)$ where $x\,(t) = 10\ \cos\ \left(8\pi \times 10^3\right)\ t$ and $T_s = 100\ \mu\,\text{sec}$. When $y\,(t)$ is passed through an ideal low-pass filter with a cutoff frequency of 5 KHz, the output of the filter is

Open solution

(A) $5 \times 10^{-6}\ \cos\ \left(8\pi \times 10^3\right)\ t$
(B) $5 \times 10^{-5}\ \cos\ \left(8\pi \times 10^3\right)\ t$
(C) $5 \times 10^{-1}\ \cos\ \left(8\pi \times 10^3\right)\ t$
(D) $10\cos\ \left(8\pi \times 10^3\right)\ t$

Q10. GATE ECE 2001

A band limited signal is sampled at the Nyquist rate. The signal can be recovered by passing the samples through

Open solution

(A) an RC filter
(B) an envelope detector
(C) a PLL
(D) an ideal low-pass filter with the appropriate bandwidth

Q11. GATE ECE 1998

Flat top sampling of low pass signals

Open solution

(A) gives rise to aperture effects
(B) implies over sampling
(C) leads to aliasing
(D) introduces delay distortion

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Q12. GATE ECE 1995

A 1.0 kHz signal is flat - top sampled at the rate of 1800 samples/sec and the samples are applied to an ideal rectangular **LPF** with cut - off frequency of 1100 Hz, then the output of the filter contains

(A) only 800 Hz component.

(B) 800 Hz and 900 Hz components.

(C) 800 Hz and 1000 Hz components.

(D) 800 Hz, 900 Hz and 1000 Hz components.

Open solution

Q13. GATE ECE 1994

Increased pulse-width in the flat-top sampling, leads to

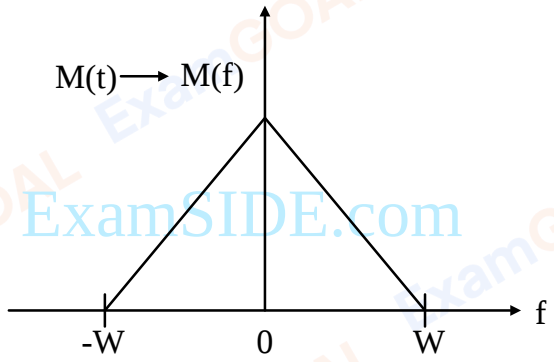


Fig. 1

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(A) attenuation of high frequencies in reproduction.

(B) attenuation of low frequencies in reproduction.

(C) greater aliasing errors in reproduction.

(D) no harmful effects in reproduction.

Open solution

Marks 2 15 questions

Q14. GATE ECE 2025

Consider a continuous-time finite-energy signal $f(t)$ whose Fourier transform vanishes outside the frequency interval $[-\omega_c, \omega_c]$, where ω_c is in rad/sec.

The signal $f(t)$ is uniformly sampled to obtain $y(t) = f(t)p(t)$. Here

$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - \tau - nT_s)$

with $\delta(t)$ being the Dirac impulse, $T_s > 0$, and $\tau > 0$. The sampled signal $y(t)$ is passed through an ideal lowpass filter $h(t) = \omega_c T_s \frac{\sin(\omega_c t)}{\pi \omega_c t}$ with cutoff frequency ω_c and passband gain T_s.

The output of the filter is given by .

(A) $f(t)$ if $T_s < \pi/\omega_c$

(B) $f(t-\tau)$ if $T_s < \pi/\omega_c$

(C) $f(t-\tau)$ if $T_s < 2\pi/\omega_c$

(D) $T_s f(t)$ if $T_s < 2\pi/\omega_c$

Open solution

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Q15. GATE ECE 2024

A continuous time signal $x(t) = 2\cos(8\pi t + \frac{\pi}{3})$ is sampled at a rate of 15 Hz. The sampled signal $x_s(t)$ when passed through an LTI system with impulse response

$h(t) = \left(\frac{\sin 2\pi t}{\pi t}\right)\cos(38\pi t - \frac{\pi}{2})$

produces an output $x_o(t)$. The expression for $x_o(t)$ is _____.

Open solution

(A) $15\sin(38\pi t + \frac{\pi}{3})$
(B) $15\sin(38\pi t - \frac{\pi}{3})$
(C) $15\cos(38\pi t - \frac{\pi}{6})$
(D) $15\cos(38\pi t + \frac{\pi}{6})$

Q16. GATE ECE 2017 Set 2

The signal x(t) = $\sin(14000\,\pi t)$, where t is in seconds, is sampled at a rate of 9000 samples per second. The sampled signal is the input to an ideal lowpass filter with frequency response H(f) as following: $H(f) = \begin{cases} 1, & |f| \leq 12\,kHz \\ 0, & |f| > 12\,kHz \end{cases}$

What is the number of sinusoids in the output and their frequency inkHz?

Open solution

(A) Number = 1, frequency = 7
(B) Number =3, frequencies =2, 7, 11
(C) Number =2, frequencies =2, 7
(D) Number =2, frequencies =7, 11

Q17. GATE ECE 2015 Set 3

Consider a continuous-time signal defined as $x(t) = \left(\frac{\sin(\pi t/2)}{(\pi t/2)}\right) * \sum_{n=-\infty}^{\infty}\delta(t-10n)$ Where ' * ' denotes the convolution operation and t is in seconds. The Nyquist sampling rate (in samples/sec) for x(t) is _____.

Open solution

Q18. GATE ECE 2010

The Nyquist sampling rate for the signal $s(t) = \frac{\sin(500\pi t)}{\pi t} \times \frac{\sin(700\pi t)}{\pi t}$ is given by

Open solution

(A) 400 Hz
(B) 600 Hz
(C) 1200 Hz
(D) 1400 Hz

Q19. GATE ECE 2006

The minimum sampling frequency (in samples /sec) required to reconstruct the following signal from its samples without distortion $x(t) = 5\left(\frac{\sin\ 2\pi 1000\,t)}{\pi t}\right)^3 + 7\left(\frac{\sin\ 2\pi 1000\,t}{\pi t}\right)^2$ would be

Open solution

(A) 2×10^3
(B) 4×10^3
(C) 6×10^3
(D) 8×10^3

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Q20. GATE ECE 2006

A signal m(t) with bandwidth 500 Hz is first multiplied by a signal g(t) where

$g(t) = \sum_{k=-\infty}^{\infty} (-1)^k \delta(t - 0.5 \times 10^{-4} k)$

The resulting signal is then passed through an ideal low pass filter with bandwidth 1 kHz. The output of the low pass filter would be

Open solution

(A) $\delta(t)$ (B) m(t) (C) 0 (D) m(t) $\delta(t)$

Q21. GATE ECE 2004

A 1 kHz sinusoidal signal is ideally sampled at 1500 samples /sec and the sampled signal is passed through an ideal low-pass filter with cut-off frequency 800 Hz. The output signal has the frequency

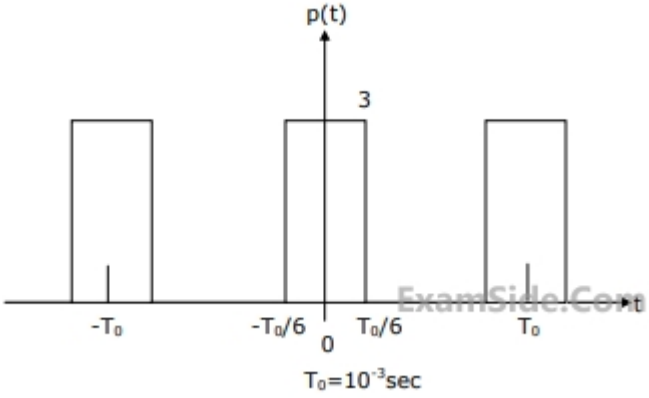
Open solution

(A) zero Hz (B) 0.75 kHz (C) 0.5 kHz (D) 0.25 kHz

Q22. GATE ECE 2003

Let x(t) = $2\cos(800\pi t) + \cos(1400\pi t)$. x(t) is sampled with the rectangular pulse train shown in figure. The only spectral components (in KHz) present in the sampled signal in the frequency range 2.5 kHz to 3.5 kHz are

Open solution



(A) 2.7, 3.4 (B) 3.3, 3.6

(C) 2.6, 2.7, 3.3, 3.4 (D) 2.7, 3.3

Q23. GATE ECE 2002

A signal x(t) = 100 cos $(24\pi \times 10^3)$ t is ideally sampled with a sampling period of 50 μ sec and then passed through an ideal low pass filter with cutoff frequency of 15 KHz. Which of the following frequencies is/ are present at the filter output?

Open solution

(A) 12 KHz only (B) 8 KHz only (C) 12 KHz and 9 KHz (D) 12 KHz and 8 KHz

Q24. GATE ECE 2001

The Nyquist sampling interval, for the signal Sinc(700t) + Sinc(500t) is

Open solution

(A) $\frac{1}{350}$ sec (B) $\frac{\pi}{350}$ sec (C) $\frac{1}{700}$ sec (D) $\frac{\pi}{350}$ sec

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Q25. GATE ECE 1999

The Nyquist sampling frequency (in Hz) of a signal given by $16 \times 10^4 \ \sin c^2(400t) * 10^6 \ \sin c^3(100t)$ is

(A) 200 (B) 300 (C) 500 (D) 1000

Open solution

Q26. GATE ECE 1991

A signal has frequency components from 300 Hz to 1.8 KHz. The minimum possible rate at which the signal has to be sampled is _____ (fill in the blank).

Open solution

Q27. GATE ECE 1990

A 4 GHz carrier is DSB-SC modulated by a low pass message signal with maximum frequency of 2 MHz. The resultant signal is to be ideally sampled. The minimum frequency of the sampling impulse train should be:

(A) 4 MHz (B) 8 MHz (C) 8 GHz (D) 8.004 GHz

Open solution

Q28. GATE ECE 1988

A signal containing only two frequency components (3 kHz and 6 kHz) is sampled at the rate of 8 kHz, and then passed through a low pass filter with a cutoff frequency of 8 kHz. The filter output

(A) is an undistorted version of the original signal.

(B) contains only the 3 kHz component.

(C) contains the 3 kHz component and a spurious component of 2 kHz.

(D) contains both the components of the original signal and two spurious components of 2 kHz and 5 kHz.

Open solution

Marks 4 1 questions

Q29. GATE ECE 1992

A low pass signal x(t) has a spectrum given by $X(f) = \begin{cases} 1 - |f|/2000, & for \ |f| \leq 2000 \, Hz \\ 0, & elsewhere \end{cases}$

Assuming that x(t) is ideally sampled at a sampling frequency of 3 kHz, sketch
(i) x(f), and
(ii) the spectrum of the sampled signal for $|f| \leq 3 \, kHz$.

Open solution

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Marks 5 2 questions

Q30. GATE ECE 2000

Open solution

A band limited signal x(t) with a spectrum X(f) as shown in Fig. a is processed as shown in Fig.b. p(t) is a periodic train of impulses as in Fig. c. The ideal band pass filter has a pass band from 26 KHz to 34 KHz.

(a) Calculate the Fourier series coefficients c_n in the Fourier expansion of p(t) in form $p(t) = \sum_{n=-\infty}^{+\infty} c_n \exp(jn2\pi t/T)$.

(b) Find the Fourier Transform of p(t).

(c) Obtain and sketch the spectrum of $x_s(t)$.

(d) Obtain and sketch the spectrum of y(t).

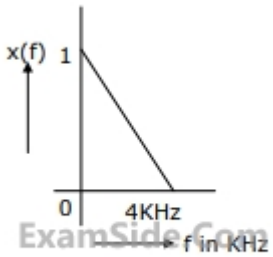


Fig. a

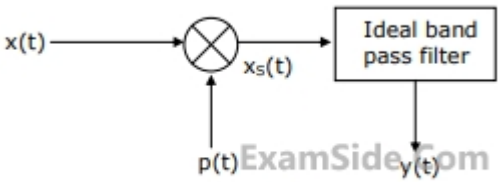


Fig. b

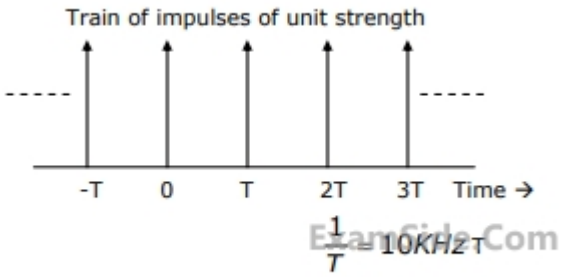


Fig.c

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Q31. GATE ECE 1993

A low pass signal m(t) band-limited to B Hz is sampled by a periodic rectangular pulse train, $p_\tau(t)$ of period T_s = 1/(3B) sec. Assuming natural sampling and that the pulse amplitude and pulse width are A volts and 1/(30B) sec, respectively, obtain an expression for the frequency spectrum of the sampled signal m_s(t)

Open solution

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Answer Key

Marks 1

1. **D**	2. **B**	3. **C**	4. **A**	5. **3**
6. **A**	7. **10**	8. **A**	9. **C**	10. **D**
11. **A**	12. **C**	13. **A**		

Marks 2

14. **A**	15. **A, C**	16. **B**	17. **0.4**	18. **C**
19. **C**	20. **C**	21. **C**	22. **D**	23. **D**
24. **C**	25. **B**	26. **3.6 KHz**	27. **B**	28. **D**

Marks 4

29. **solve it.**	

Marks 5

30. **solve it**	31. $M_s(f) = 0.1A \sum_{n=-\infty}^{\infty} \sin c(0.1n) M(f - nf_s)$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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